

Network Security

Module 02

Duration: 24:29

Video: <https://www.youtube.com/watch?v=kibMPvGrLXw>

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0:04 hello everyone I am pradeep bandari from duaba college jander I'm going to present a session on network security

0:12 over the past few years the internet enabled business or E business has drastically improved company's

0:20 efficiency and revenue growth e business applications such as e-commerce Supply Chain management and

0:29 remote assess enable companies to streamline processes lower operating costs and increase customer

0:37 satisfaction such applications require Mission critical networks that accommodate voice video and data traffic

0:45 and these networks must be scalable to support increasing number of users and the need for greater capacity and

0:54 performance however as networks enable more and more applications and are available to more and more users they

1:02 become ever more vulnerable to a wider range of security threats to combat these threats and

1:11 ensure that e business transactions are not compromised network security play a

1:17 major role in today's networks network security deals with securing the information through cryptography and

1:26 network security devices the main goal of net network security confidentiality now confidentiality is

1:34 probably the most common aspect of information security we need to protect our confidential information an

1:42 organization needs to guard against those malicious actions that endanger the confidentiality of its information

1:51 confidentiality not only applies to the storage of the information it also applies to the transmission of information

2:00 when we send a piece of information to be stored in a remote computer or when we retrieve a piece of information from

2:09 a remote computer we need to conceal it during transmission second is integrity
2:17 information needs to be changed constantly in a bank when a customer deposits or
withdraws money the balance
2:26 of her account needs to be changed in integrity means that changes need to be
done only by authorized entities and
2:36 through authorized mechanisms Integrity violation is not necessarily the result
2:43 of a malicious act and interruption in the system such as a power failure or a power
surge may also create unwanted
2:52 changes in some information third is availability the third component of information
security
3:00 the information created and stored by an organization needs to be available to
authorized entities information is
3:09 useless if it is not available information needs to be constantly changed which
means it must be
3:16 accessible to authorized entities the unavailability of information is just as harmful
for an organization as the lack
3:25 of confidentiality or Integrity imagine what would happen happen to a bank if the
customers could not access their
3:33 accounts for the transaction security attacks now any action that compromises the
security of
3:40 information owned by an organization is called security attack two things to be
considered separately in context of
3:49 security attacks are threat a potential for violation of security which exists
3:57 when there is a circumstance capability action or event that could breach security
and cause harm that is a threat
4:05 is a possible danger that might exploit a vulnerability second is attack an
4:12 assault on system security that derives from an intelligent threat that is an
intelligent act that is a deliberate
4:21 attempt to evade Security Services and violate the security policy of a system now
security attack attacks can be of
4:30 two types these are passive attacks and active attacks passive attacks passive
4:36 attacks are in nature of if dropping on or monitoring of transmission the goal
4:44 of the opponent is to obtain information that is being transmitted two types of
passive attacks are release of message
4:53 contents and traffic analysis release of message contents is easily understood a
5:00 telephone conversation an electronic message and a transferred file may contain
sensitive or confidential
5:09 information we would like to prevent an opponent from learning the contents of
these

5:16 Transmissions the figure on the screen reveals the concept of if dropping a second type of passive attack

5:25 is traffic analysis suppose that we have had a way of masking the contents of messages or

5:33 other information traffic so that the opponents even if they captured the message could not extract the

5:41 information from the message the common technique for masking contents is encryption if we had encryption

5:49 protection in place an opponent might still be able to observe the pattern of these messages the opponent could

5:57 determine the location and identity of communicating hosts and could observe the frequency and length of messages

6:05 being exchanged the information might be useful in guessing the nature of the communication that was taking place the

6:15 figure on the screen reveals the concept of traffic analysis replay involves the

6:22 passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect as shown on the

6:31 screen modification of messages simply means that some portion of a legitimate message is altered or that messages are

6:41 delayed or reordered to produce an unauthorized effect for example a

6:47 message meaning allow John Smith to read confidential file accounts is modified

6:54 to mean allow Fred Brown to read confidential file

7:01 7 minutes, 1 second counts the concept is shown on the screen the denial of service prevents or

7:08 inhibits the normal use or management of communication facilities this attack may

7:16 have a specific Target for example an entity May suppress all messages directed to a particular

7:25 destination another form of service denial is the disruption of an entire network either by disabling the network

7:35 or by overloading it with messages so as to degrade performance the concept is shown on the

7:44 screen are three goals of security that is confidentiality integrity and

7:51 availability can be threatened by security attacks we can divide security attacks

7:58 into three groups related to the security goals as well A New Concept snooping is introduced in this

8:07 classification snooping means unauthorized access to or interception

8:13 of data for example a file transferred through the internet may contain confidential information an unauthorized

8:22 entity May intercept the transmission and use the content for her own benefit to

prevent grouping the data can be made

8:31 meaningless for the Interceptor by using encryption techniques let us discuss various

8:37 network security service techniques some security services are required to achieve security goals and prevent

8:45 attacks the actual implementation of security goals needs two techniques these techniques are designed

8:54 to protect one or more attacks while maintaining security goals the the first technique is cryptography and second is

9:02 security through specific devices cryptography a word with Greek origin means secret

9:09 writing however we use the term to refer to the science and art of transforming messages to make them secure and immune

9:19 to attacks although in the past cryptography referred only to the encryption and decryption of messages

9:26 using secret keys today is defined Ed as involving three distinct mechanisms

9:33 these are symmetric key cryptography asymmetric key cryptography and hashing let us discuss symmetric key

9:41 cryptography it is also known as traditional Cipher in symmetric key cryptography same key is used for encryption and

9:50 decryption and the key can be used for bidirectional communication the figure on the screen represents the general concept of

9:59 symmetric key cryptography in this model plain text is used as input we convert the plain text

10:06 into Cipher text by using an encryption algorithm and key on the receiver side Cipher text is converted into plain text

10:14 by using decryption algorithm and key in symmetric cryptography sender and receiver share the same secret key

10:23 asymmetric algorithm rely on one key for encryption and a different but Rel Ed key for decryption these algorithms have

10:33 the following important characteristic it is computationally invisible to determine the decryption key given only

10:41 knowledge of the cryptographic algorithm and the encryption key in addition some algorithms such as RSA also exhibit the

10:51 following characteristics either of the two related keys can be used for encryption with the other used for decryption the

10:59 main components of asymmetric key cryptography are as shown on the screen plain text this is the readable message

11:08 or data that is fed into the algorithm as input encryption algorithm performs various

11:16 Transformations on the plain text public and private keys this is a pair of keys that
have been selected so that if one
11:24 is used as encryption key the other is used as decryption key the exact
Transformations performed by the
11:33 algorithm depend on the public or private key that is provided as input Cipher text
this is The Scrambled
11:43 message produced as output it depends on the plain text and the key for a given
message two different keys will produce
11:52 two different Cipher texts decryption algorithm this algorithm accepts the
11:58 cipher for text and the matching key and produces the original plain text hash
12:05 functions a cryptographic hash function takes a message of arbitrary length and
12:11 creates a message digest of fixed length all cryptographic hash functions need to
12:19 create a fixed size digest out of a variable size message creating such a function is
best accomplished using
12:28 iteration instead of using a hash function with variable size input a function with
fixed size input is
12:36 created and is used as necessary number of times the fixed size input function
12:44 is referred to as a compression function it compresses an nbit string to create
12:52 an M bit string where n is normally greater than M the scheme is referred to
12:59 as has an iterated cryptographic hash function several hash algorithms were
designed by Ron rivest these are
13:08 referred to as md2 md4 md5 where MD stands for message digest the last
13:15 version md5 is a strengthened version of md4 that divides the message into blocks
13:22 of 512 bits and creates a 128bit Digest it turns out that a
13:30 message digest of size 128 bits is too small to resist attack the secure hash
13:39 algorithm sha is a standard that was developed by the National Institute of
13:45 Standards and Technology called nist sha has gone through several versions now
13:52 let's discuss the network security through specific network devices when we think
about network security we can also
14:01 14 minutes, 1 second think about the boundary in which our network is established
we need to protect our network from external
14:09 attacks and we also need to mind internal suspicious activities we usually deploy
security devices as shown
14:19 in the screen to protect our Network border routers as static filtering
14:26 device firewalls in in detection systems idss IPS that is intrusion prevention
14:34 systems border routers as static filtering devices now routers are the
14:41 traffic cops of the network they direct traffic into out of and within our
14:48 networks as you can observe in the figure the Border router is the last router you
control before an untrusted

14:56 Network such as the internet because all of internet traffic goes through this
15:03 router it often functions as in Network's first and last line of defense
15:10 through initial and final filtering we can configure border router as static packet
filtering device by
15:19 configuring ACL that is access control lists we called router as static filtering
device because it can inspect
15:28 packets only up to layer three that is Network layer improperly destined traffic
might be internal addresses that
15:37 hit your external interface or vice versa and they can be addressed with Ingress
and egress
15:46 filtering border routers can also block traffic that is considered high risk
15:52 from entering your network icmp is a favorite of attackers both for denial of
15:59 service attack and Denial of Service so blocking this protocol in whole or in part is a
common function of
16:07 a border router you may also consider blocking Source routed packets at the
border
16:14 router because they can circumvent defenses the Border router can also Block
out of band packets such as syn
16:23 packets on February 9 2000 websites such as Yahoo and CNN were temporarily
taken
16:30 off the internet mostly by distributed denial of service Smurf attacks a Smurf
16:37 attack involves sending spoofed icmp Echo request that is pings to the broadcast
16:44 addresses resulting in a response from every host in this case the spoofing
16:51 allowed attackers to direct the large number of responses to a victim Network
Ingress and egress filtering would have
17:00 blocked the spoofed traffic and allowed them to weather the distributed denial of
service
17:08 so every Network should have Ingress and Egress filtering at the border router
17:15 to permit only traffic that is destined for the internal Network to enter and traffic
that is destined for the
17:23 external network to exit Firewall a firewall is a choke point
17:30 device that has a set of rules specifying what traffic it will allow or deny to pass
through it a firewall
17:38 typically picks up where the Border router leaves off and makes a much more
thorough pass at filtering traffic
17:48 firewalls can be of three types static stateful and proxy Firewalls although
17:56 firewalls aren't perfect effect they do block what we tell them to block and allow
what we tell them to allow as
18:05 previously discussed router can be used as static packet filter or as static

18:13 firewall but we need specialized firewalls to implement as stateful and
18:20 proxy firewalls unlike static packet filtering devices stateful firewalls
18:27 keep track of Connections in a state table and are the most common type of
18:34 firewalls a stateful firewall blocks traffic that is not in its table of
18:40 established connections the firewall rule base determines the source and
destination IP and port numbers
18:49 permitted to establish connections by rejecting non-established non-permitted
18:56 connections a stateful fireball helps to block reconnaissance packets as well as
those that may gain more extensive
19:05 unauthorized access to protected resources proxy firewalls are the most advanced
and least common type of
19:14 firewalls proxy firewalls are also stateful in that they block any
19:21 non-established non-permitted connections as with stateful firewalls the firewall
rule base determines the
19:30 source and destination IP and port numbers that are permitted to establish
connections proxy firewalls offer a high
19:39 level of security because internal and external hosts never communicate directly
rather the fireball acts as an
19:49 intermediary between hosts proxy firewalls examine the entire packet to ensure
compliance with the protocol that
19:58 is indicated by the destination port number ensuring that only protocol
compliant traffic passes through the
20:06 fire wall helps defense in depth by diminishing the possibility of malicious traffic
entering or exiting from your
20:15 network proxy firewalls diminishes the possibility of malicious traffic entering or
exiting your network by
20:23 ensuring that only protocol compliant traffic passes through if intrusion detection
systems now an IDS is like a
20:32 security alarm system for your network that is used to detect an alert on malicious
20:40 events the system might comprise many different IDS sensors placed at strategic
points in your network in
20:49 general IDS sensors watch for predefined signatures of malicious events and they
20:56 might perform statistical and anomaly analysis when IDS sensors detect
suspicious events they can Alert in
21:05 several different ways including email paging or simply logging the
21:12 occurrence a network IDS could identify and alert on the following first is DNS
Zone transfer
21:20 request from unauthorized host unicode attacks directed at a web server buffer
overflow attack

21:29 V propagation Etc intrusion prevention systems now an IPS is a system that
21:39 automatically detects and prevents Network and hosts attacks in contrast to
21:46 a traditional IDs which focuses on notifying the administrator of anomalies
21:52 and IPS strives to automatically defend the target without administrators direct
21:59 involvement protection may involve using signature-based or behavioral techniques
to identify an attack and then blocking
22:09 the malicious traffic or system call before it causes harm in this respect an
22:16 IBS combines the functionality of a firewall and IDs to offer a solution
22:23 that automatically blocks offending actions as soon as it detects and attack
frequently we get caught up in the
22:32 technical aspect of network security without considering its non-technical elements
tasks such as optimizing the
22:41 firewall rule base examining Network traffic for suspicious patterns and locking
down the configuration of
22:49 systems are certainly important to network security what we often forget is the
human end of
22:57 things such as has policies and awareness that go along with the Technical
Solutions policy determines what
23:06 security measures your organization should Implement as a result the security
policy guides your decisions
23:15 when implementing security of the network an effective defense and depth
infrastructure requires a comprehensive
23:23 and realistic security policy Hallmarks of good security policy
23:30 include Authority who is responsible scope who it affects expiration when it
23:38 ends specificity what is required and Clarity can everyone understand it so in
23:46 this session we have discussed what is network security security issues and basic
network security techniques I hope
23:56 you have enjoyed the session thank you so much much
24:01 24 minutes, 1 second [Music]
24:20 [Music]